

FROM THE SEQUENCE TO THE SPECTRUM: THE OSCILLATION OF μ_n FOR THE $h = 2$ FAMILY, VIA THEOREM 2.3

BARRY BRENT

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This note carries the analysis of the constant sequence $h_0 = 1$, $h_n = 2$ ($n \geq 1$) one step further than the two companion documents, which stopped at the coefficients h_n (undeformed) and $h_n^{(c)}$ (deformed, with the limits $\cosh c, \sinh c$). Here we pass all the way to the spectrum, i.e. to the minimum-modulus eigenvalue $\mu_n = \min_{v \in E(\chi_n)} |v|$ of the matrix $J_{h,n}$, the quantity actually plotted in the manuscript. As requested, we do *not* go through the determinant/ χ construction: we use Theorem 2.3 of the manuscript to pass directly from the sequence to the characteristic polynomial, and then read the spectrum off the result.

The outcome is a complete and fully explicit account of the oscillation seen in Figure 48B, including a correction to the unverified remark recorded there. In one sentence: *deformation by c rigidly translates the entire eigenvalue cloud by c , and the apparent oscillation of μ_n is nothing but the parity alternation of the truncated cosh and sinh polynomials — driven by exactly the same $(-1)^n$, coming from the pole at $t = -1$, that splits the coefficients into $\cosh c$ and $\sinh c$.*

1. THE OPERATOR BRIDGE

We write $D = \frac{d}{dx}$ and recall that $\chi_n(x) := \chi(J_n(\bar{j}))(x)$ denotes the characteristic polynomial attached to the sequence h through equation (D).

Proposition 1.1. *Let $H(t) = \sum_{r \geq 0} h_r t^r$ be the generating function of the sequence. Then*

$$\chi_n(x) = H(-D) x^n,$$

the differential operator $H(-D)$ (a polynomial in D when applied to x^n , since $D^r x^n = 0$ for $r > n$) applied to the monomial x^n .

Proof. Theorem 2.3 of the manuscript states

$$\chi_n(x) = \sum_{r=0}^n \binom{n}{r} r! h_r (-1)^r x^{n-r}.$$

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Since $\binom{n}{r}r! = \frac{n!}{(n-r)!}$ and $D^r x^n = \frac{n!}{(n-r)!} x^{n-r}$, each summand is $(-1)^r h_r D^r x^n$, so $\chi_n(x) = (\sum_{r \geq 0} h_r (-D)^r) x^n = H(-D) x^n$. $\square$

Thus every structural feature of H — its poles, its residues, an exponential factor — becomes an operator acting on x^n , and the spectrum of J_n is the zero set of the resulting polynomial. Two facts about $H(-D)$ will do all the work: a simple pole of H at $t = t_0$ contributes the operator $\frac{1}{1 - t_0^{-1}(-D)}$ type resolvent $\frac{\rho}{t_0 - (-D)}$, and the entire factor e^{ct} contributes e^{-cD} , which is the translation operator $(e^{-cD} f)(x) = f(x - c)$.

2. THE DEFORMED FAMILY: A RIGID SHIFT AND A cosh / sinh DICHOTOMY

For the deformed sequence $H^{(c)}(t) = e^{ct}/(1 - t^2)$, Proposition 1.1 and the partial fraction $\frac{1}{1 - t^2} = \frac{1}{2}(\frac{1}{1-t} + \frac{1}{1+t})$ give

$$\chi_n^{(c)}(x) = H^{(c)}(-D) x^n = e^{-cD} \frac{1}{1 - D^2} x^n = \left(\frac{1}{1 - D^2} y^n \right) \Big|_{y=x-c}.$$

Write $Q_n(y) := \frac{1}{1 - D^2} y^n$, the polynomial solution of $Q_n - Q_n'' = y^n$, namely $Q_n = \sum_{k \geq 0} D^{2k} y^n$.

Lemma 2.1.

$$Q_n(y) = \frac{n!}{2} (E_n(y) + (-1)^n E_n(-y)) = \begin{cases} n! \sum_{0 \leq 2i \leq n} \frac{y^{2i}}{(2i)!}, & n \text{ even (a truncated cosh)}, \\ n! \sum_{0 \leq 2i+1 \leq n} \frac{y^{2i+1}}{(2i+1)!}, & n \text{ odd (a truncated sinh)}, \end{cases}$$

where $E_n(z) = \sum_{m=0}^n z^m/m!$ is the n -th Taylor polynomial of $\exp$.

Proof. $\frac{1}{1-D} y^n = \sum_{k \geq 0} D^k y^n = n! \sum_{m=0}^n y^m/m! = n! E_n(y)$, and likewise $\frac{1}{1+D} y^n = (-1)^n n! E_n(-y)$. Average the two (the partial fraction). Finally $E_n(y) + E_n(-y)$ retains the even powers and $E_n(y) - E_n(-y)$ the odd ones. $\square$

Corollary 2.2 (deformation is a rigid translation). $\chi_n^{(c)}(x) = Q_n(x - c)$, hence

$$\text{spec}(J_n^{(c)}) = c + \text{spec}(J_n^{(0)}), \quad \mu_n^{(c)} = \min_{Q_n(y)=0} |c + y|.$$

The parameter c translates the whole eigenvalue cloud by c and changes nothing else.

Theorem 2.3. Fix $c \in \mathbb{R}$. Then

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ even}}} \mu_n^{(c)} = \sqrt{c^2 + \frac{\pi^2}{4}}, \quad \mu_n^{(c)} = |c| \text{ for every odd } n \text{ (for all } n \text{ large enough this is the exact value)}.$$

Proof. By Lemma 2.1, $\frac{1}{n!}Q_n(y) \rightarrow \cosh y$ (even n) and $\rightarrow \sinh y$ (odd n), uniformly on compact sets, because the partial sums $E_n(\pm y) \rightarrow e^{\pm y}$ do. By Hurwitz's theorem the zeros of Q_n in any fixed disc converge to those of the limit function; the remaining (“spurious”) zeros escape to infinity, their moduli growing linearly in n . The zeros of $\cosh$ are $\pm i\pi(k + \frac{1}{2})$, the smallest being $\pm i\pi/2$; the zeros of $\sinh$ are $\pm i\pi k$, namely 0 and then $\pm i\pi, \dots$

For even n the smallest-modulus zeros of Q_n approach $\pm i\pi/2$, so by Corollary 2.2 the eigenvalue nearest the origin approaches $c \pm i\pi/2$ and $\mu_n^{(c)} \rightarrow |c \pm i\pi/2| = \sqrt{c^2 + \pi^2/4}$.

For odd n , Q_n is an odd polynomial, so $y = 0$ is a zero *exactly*; hence $x = c$ is an eigenvalue of $J_n^{(c)}$ for every odd n , giving $\mu_n^{(c)} \leq |c|$. The remaining zeros approach $\pm i\pi, \pm 2i\pi, \dots$ (moduli $\geq \pi$) together with the spurious zeros (moduli $\rightarrow \infty$); once n is large enough that all nonzero zeros y satisfy $|c + y| > |c|$, the minimum is attained at $y = 0$ and $\mu_n^{(c)} = |c|$. $\square$

Remark 1 (the case $c = 1$, and a correction to the manuscript). For $c = 1$ the theorem gives $\mu_n^{(1)} = 1$ on odd n and $\mu_n^{(1)} \rightarrow \sqrt{1 + \pi^2/4} = 1.86209588911\dots$ on even n . High-precision root-finding via Theorem 2.3 confirms this to many digits and shows the odd value is 1 *exactly* for every odd n in the computed range (see Figure 1, left, and Figure 2).

This settles the unverified remark at Figure 48B. The even limit $\sqrt{\pi^2/4 + 1}$ recorded there is correct. The companion statement that “the heights of the odd sequence really are exactly zero” is *not* correct: a height $\mu_n = 0$ would require $0 \in \text{spec}(J_n^{(c)})$, i.e. $\chi_n^{(c)}(0) = (-1)^n n! h_n^{(c)} = 0$, i.e. $h_n^{(c)} = 0$; but $h_n^{(c)} \rightarrow \sinh c = \sinh 1 \neq 0$ on odd n , so 0 is never an eigenvalue for large odd n . The correct odd height is the *shifted origin* $x = c$, of modulus $|c| = 1$ — exactly the “apparently constant value one” that the manuscript reports from the plot.

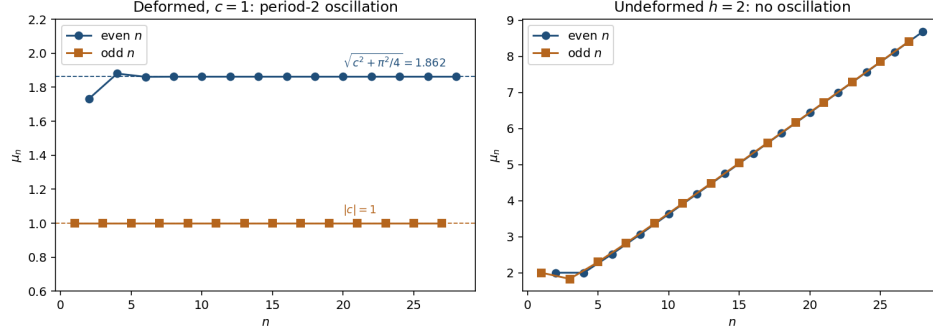


FIGURE 1. Minimum-modulus eigenvalue μ_n computed from Theorem 2.3. *Left*: deformed $c = 1$. Odd n sit on $|c| = 1$ (the eigenvalue $x = c$ forced by the zero of the truncated sinh); even n climb to $\sqrt{c^2 + \pi^2/4} = 1.862$ (the shifted zero $c \pm i\pi/2$ of the truncated cosh). This period-2 split *is* the oscillation. *Right*: undeformed $h = 2$. The even and odd points lie on a single smooth curve growing linearly — no parity split, no oscillation.

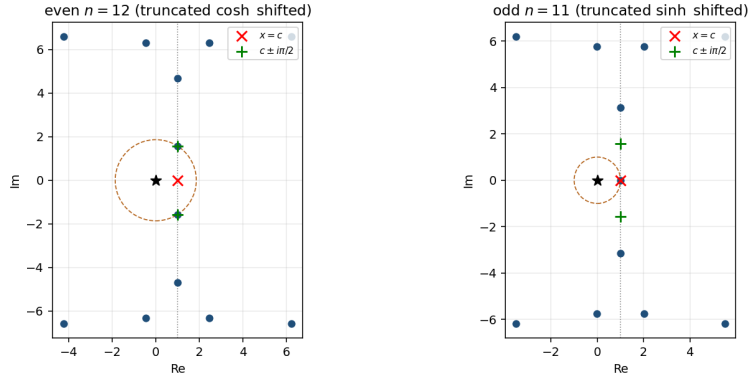


FIGURE 2. Eigenvalue clouds of $J_n^{(1)}$ in $\mathbb{C}$. By Corollary 2.2 they are the zeros of a truncated cosh (even n , left) or sinh (odd n , right), rigidly shifted to the line $\text{Re} = c = 1$ (dotted). The star marks the origin and the dashed circle has radius μ_n . *Left*: the nearest eigenvalues are $c \pm i\pi/2$ (green +). *Right*: the truncated sinh vanishes at 0, so $x = c$ (red x) is an eigenvalue and is the nearest one.

3. THE UNDEFORMED SEQUENCE: ONE TRUNCATED EXPONENTIAL, NO DICHOTOMY

For the undeformed sequence $H(t) = (1+t)/(1-t)$ the companion note records the boundary decomposition $H(t) = \frac{2}{1-t} - 1$: a single pole, at $t = +1$, with the pole at $t = -1$ *absent*. Proposition 1.1 turns this directly into

$$\chi_n(x) = H(-D) x^n = \left(\frac{2}{1+D} - 1 \right) x^n = 2(-1)^n n! E_n(-x) - x^n.$$

The transcendental object controlling the zeros is now the truncated *exponential* $E_n(-x)$ — the *same* function for both parities; there is no cosh / sinh alternation, because there is no second pole to supply the $\frac{1}{1-D}$ summand. The correction $-x^n$ is negligible near the zeros of $E_n(-x)$: at such a zero $|x|$ grows like κn (Szegő: x/n accumulates on the curve $|x e^{1-x}| = 1$, $|x| \leq 1$, with $\kappa \approx 0.278$), and the ratio of the two terms is $\sim (\kappa e)^n / (2\sqrt{2\pi n}) \rightarrow 0$. Hence the smallest-modulus eigenvalue grows linearly, $\mu_n \sim \kappa n \rightarrow \infty$, as a single smooth (eventually increasing) sequence with no even/odd split (Figure 1, right). In particular 0 is never an eigenvalue, since $\chi_n(0) = (-1)^n \cdot 2 \cdot n! \neq 0$.

4. WHAT MAKES THE DIFFERENCE

Remark 2 (the pole at $t = -1$ is the whole story, at the spectral level too). Comparing the two computations, the oscillation of μ_n is governed by precisely one feature of H : whether it has a second boundary pole, at $t = -1$.

- Under $t \mapsto -D$, a pole at $t = +1$ becomes the operator $\frac{1}{1+D}$ (truncated exponential), and a pole at $t = -1$ becomes $\frac{1}{1-D}$. The deformed $1/(1-t^2)$ has *both*, with *equal* residues, so they combine into the symmetric pair $E_n(y) \pm E_n(-y)$ — the truncated cosh and sinh of Lemma 2.1. Their alternation with the parity of n is the spectral oscillation, and the alternating sign is the very $(-1)^n$ that, at the coefficient level, splits $h_n^{(c)}$ into $\cosh c$ and $\sinh c$.
- The undeformed $(1+t)/(1-t)$ cancels the $t = -1$ pole with its numerator zero (companion note, Lemma 2.1), leaving only $\frac{1}{1+D}$ — one truncated exponential, no alternation, no oscillation.

Thus the coefficient oscillation and the spectral oscillation are not two phenomena but one, with a single cause (the pole at $t = -1$); Theorem 2.3, in the form $\chi_n = H(-D)x^n$, is the exact conduit that carries that cause from the sequence to the spectrum. A pair of boundary poles at $e^{\pm i\theta}$ would, by the same mechanism, produce eigenvalues near $c \pm i\theta$ -scaled directions and an oscillation of angular frequency θ ; the case $\theta = \pi$ (poles at ± 1) is the period-2 even/odd split seen here.

Remark 3 (scope). Everything above is exact for the $h = 2$ family, where H is rational and $H(-D)$ is a genuinely finite/resolvent operator. For the modular-form sequences that motivate the manuscript the generating functions are not rational, the operator $H(-D)$ is not a finite combination of resolvents, and locating the boundary singularities is itself the hard problem; the present analysis does not claim to explain those cases. What it does provide is a fully worked mechanism in the one family where every step is explicit, and a precise template for the rest: find the dominant boundary singularities of H , and read the oscillation — in the coefficients and, through $\chi_n = H(-D)x^n$, in the spectrum — off their locations.

Email address: `axcjh@bu.edu`